

WASSIM (WES) BOUAZIZ

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RESEARCH EXPERIENCE

Meta & École polytechnique, *Research Assistant, PhD student*

2022 – 2025

FAIR & CMAP department at École polytechnique

Paris, France

I explored a spectrum research questions at the intersection of AI Security, Safety, and Interpretability.

- Published research work (either as first author or contributor) in top venues including ICLR, ICASSP and NeurIPS.
- Contributed to Llama 3 models by developing safety evaluations and safety data annotation campaigns.

Independant, *Machine Learning Consultant*

Jan. 2021 – May. 2022

Consulting and Teaching in Machine Learning

Paris, France

- Giving Deep Learning lectures and supervising projects at Le Wagon.
- Advising startups on project feasibility and state of the art in machine learning research on various topics.

Facebook AI Research, *Research Intern*

Jul. – Nov. 2020

Noise-robust Automatic Speech Recognition

Palaiseau, France

- Achieved 10% performance improvement using audio data augmentation to enhance robustness of speech recognition systems.

ENS, *Research Intern*

Apr. – Aug. 2019

ENS CoML team

Paris, France

- Classification of child/adult directed speech for a world-wide study on child language acquisition (ACLEW).

CentraleSupélec, *Graduate Research Assistant*

Oct. 2018 – Apr. 2019

Graphs, Algorithms and Combinatorics team

Paris, France

- Contributed to improving the known time complexity of minimum cuts counting in graphs.

Snips (Acquired by Sonos), *Embedded Software Intern - AI*

Jun. – Aug. 2018

Embedded systems - Real time inference

Paris, France

- Developed real time inference application using Snips' voice assistant with a $\leq 30\%$ CPU usage on a Cortex-M7 microcontroller.

EDUCATION

Meta & École polytechnique, *PhD in Mathematics and Computer Science*

2022 – present

Fairness, Game Theory, and AI

Paris, France

Academic advisor: El Mahdi El Mhamdi (CMAP, École polytechnique)

Company advisor: Nicolas Usunier (Helsing), Max Nickel (FAIR, Meta)

Paris-Saclay University, *MSc in Mathematics, Vision and Learning ("MVA")*

2019 – 2020

Among the World's best university in Mathematics in Shanghai's ranking - with Highest Honors, GPA: 4.0/4.0

Paris, France

Université Paris-Saclay, *MSc degree in Computer Science and Machine Learning*

2018 – 2019

Double degree with CentraleSupélec, GPA: 4.0/4.0

Paris, France

CentraleSupélec, *MEng degree in Mathematics and Computer Science.*

2016 – 2019

Top Engineering school in Computer Science and Electrical Engineering. GPA: 3.9/4.0

Gif-sur-Yvette, France

Preparatory classes, *Lycée Blaise Pascal*

2014 – 2016

Two-year undergraduate intensive program for the entrance exams to top French Engineering schools

Orsay, France

TECHNICAL SKILLS

AI Large scale training, Language models training (pre & post training),

Programming Python (PyTorch), Bash, Latex, C++⁽⁻⁾ SQL⁽⁻⁾, HTML/CSS/JS⁽⁻⁾,

Languages French (*native*), English (*C1*), Spanish (*B2*), Arabic (*A2*), Japanese (*A1*)

EXTERNAL ENGAGEMENT

Reviewer ICLR (2023, 2025), NeurIPS (2024)

Invited Talks

- "Harnessing AI's Fragility: A Tool for Dataset Tracing" in Purrlab seminar (2025)
- "Harnessing AI's Fragility: A Tool for Dataset Tracing" in INRIA Sierra seminar (2025)
- "AI Security & Safety" in the "Reliable and Collaborative learning" course at École polytechnique (2024)
- "Digital commons in AI research" in the "Introduction to digital commons and public policy making" course for Public Policy Master students at Sciences Po Paris (2023, 2024)

PUBLICATIONS

– First author publications

Winter Soldier: Hypnotizing Language Models at Pre-Training with Indirect Data Poisoning

Wassim (Wes) Bouaziz, Mathurin Videau, Nicolas Usunier, El Mahdi El Mhamdi

April 2025

1st GenAI Watermarking
Workshop at ICLR 2025

Data Taggants: Dataset Ownership Verification via Harmless Targeted Data Poisoning,

Wassim (Wes) Bouaziz, Nicolas Usunier, El Mahdi El Mhamdi

October 2024

ICLR 2025

Targeted Data Poisoning for Black-Box Audio Datasets Ownership Verification,

Wassim (Wes) Bouaziz, El Mahdi El Mhamdi, Nicolas Usunier

October 2024

ICASSP 2025

Inverting Gradient Attacks Makes Powerful Data Poisoning,

Wassim (Wes) Bouaziz, El Mahdi El Mhamdi, Nicolas Usunier

October 2024

arXiv

Clustering Head: A Visual Case Study of the Training Dynamics in Transformers,

Ambroise Odonnat, Wassim (Wes) Bouaziz, Vivien Cabannes

December 2024

arXiv

Easing Optimization Paths: a Circuit Perspective,

Ambroise Odonnat, Wassim (Wes) Bouaziz, Vivien Cabannes

December 2024

Lecture ICASSP 2025

– Collaboration publications

On Monotonicity in AI Alignment

Gilles Bareilles, Wassim (Wes) Bouaziz, Sadegh Farhadkhani, Julien Fageot, Lê-Nguyen Hoang, Sébastien Rouault, Peva Blanchard, El-Mahdi El-Mhamdi

May 2025

Under review

Cultivating Pluralism In Algorithmic Monoculture:

The Community Alignment Dataset

Lily H Zhang, Smitha Milli, Karen Long Jusko, Jonathan Smith, Brandon Amos,

Wassim (Wes) Bouaziz, Jack Kussman, Manon Revel, Lisa Titus, Bhaktipriya Radharapu, Jane Yu, Vidya Sarma, Kristopher Rose, Maximilian Nickel

May 2025

Under review

The Llama 3 Herd of Models,

Llama Team, AI @ Meta

July 2024

arXiv

Iteration head: A mechanistic study of chain-of-thought,

Vivien Cabannes, Charles Arnal, Wassim Bouaziz, Xingyu Yang, Francois Charton, Julia Kempe

December 2024

NeurIPS 2024

On the Parameterized Complexity of Counting Small-Sized Minimum (S, T) -Cuts,

Pierre Bergé, Wassim Bouaziz, Arpad Rimmel, Joanna Tomasik

December 2023

SIAM Journal on Disc. Math.

Pyannote.audio: neural building blocks for speaker diarization,

Hervé Bredin, et al.

May 2020

ICASSP 2020